Sri Harsha Mudumba

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SUMMARY

Software Engineer with 3+ years of experience working at the intersection of machine learning and systems engineering. Specialized in deploying and optimizing ML inference pipelines with a focus on latency, throughput, and reliability. Strong background in performance profiling, data pipelines, and hardware-aware system design for scalable AI workloads.

TECHNICAL SKILLS

Programming & Data Processing: Python, C++ (modern standards), SQL, Bash, Object-Oriented Programming, Data Structures and Algorithms, Linux Systems.
ML Inference & Optimization: Inference optimization, latency and throughput profiling, memory efficiency, early-exit architectures, quantization (post-training and inference-aware), sparsity-aware execution, model benchmarking, performance validation.
Machine Learning Frameworks: PyTorch, TensorFlow, Hugging Face, ONNX Runtime, IREE.
Natural Language Processing & GenAI: Text Preprocessing, Tokenization, Named Entity Recognition (NER), Topic Modeling, Sentiment Analysis, Transformers, Hugging Face, LangChain Retrieval-Augmented Generation (RAG), Vector Databases (FAISS, Pinecone)
Deployment & ML Systems: FastAPI, REST-based model serving, batch and real-time inference pipelines, experiment tracking (MLflow), model versioning, CI/CD for ML systems, Docker, Kubernetes (basic)
Cloud & Data Engineering: AWS (S3, EC2, Lambda, SageMaker), Azure Machine Learning, ETL Pipelines, Distributed Training, Batch & Real-Time Inference Optimization.

RESEARCH EXPERIENCE

DURACIM – RL-Guided Early-Exit Co-Design for Compute-in-Memory Systems | Reinforcement Learning, Python, PyTorch
• Designed reinforcement learning agents (PPO, Q-learning) to optimize sequential decision policies under accuracy, latency, energy, and system endurance constraints.
• Built a simulation-driven RL pipeline integrating PyTorch execution with the CIMLOOP, a cycle-accurate simulator to model per-layer energy and memory degradation.
• Achieved 86–88% relative inference energy savings on ResNet-50 compared to full-depth execution while preserving accuracy within approximately one percentage point.
• Increased the effective RRAM lifetime to 1.02 × 10^7 years under the endurance model even in all-RRAM configurations, PPO-discovered policies extended the lifetime by 1.9 times.

PROFESSIONAL EXPERIENCE

Software Engineer, *Cognizant Technology Solutions* *Aug 2020 – Jul 2023 |KAR, INDIA*
• Profiled and optimized latency-critical production systems using Python profiling tools and Bash scripts, analyzed execution paths, memory behavior, and I/O contention, which lowered average request latency and improved overall throughput.
• Debugged reliability and performance issues in distributed inference pipelines under strict SLAs by performing root-cause analysis and failure isolation with Bash logging and monitoring tools, which restored 99.9% service availability and met the response-time target.
• Collaborated with application and infrastructure teams to evaluate compute, memory, and responsiveness trade-offs in high-throughput services, leveraging vector databases (FAISS, Pinecone) and object-oriented design, leading to a more balanced resource allocation and smoother scaling.

PROJECTS

TorchWeave-LLM – Continuous Batching Inference Server | Python, PyTorch, FastAPI, Docker **(Link)**
• Designed a continuous batching inference server with asynchronous scheduling to maximize GPU utilization under concurrent request load, achieving 4× throughput improvement.
• Reduced end-to-end inference latency by 35% by minimizing padding overhead and avoiding head-of-line blocking common in static batching approaches.
• Benchmarked latency, throughput, and GPU utilization across varying batch sizes and request arrival patterns, analyzing trade-offs between tail latency and accelerator efficiency in real-time inference systems.
• Profiled inference execution using latency and GPU utilization metrics to identify batching inefficiencies, guiding scheduling decisions that balance tail latency against accelerator efficiency in real-time workloads.
ARGUS – Multimodal Retrieval-Augmented Generation (RAG) System | *L*angChain, LangGraph, OpenAI, FAISS **(Link)**
• Built a hybrid RAG framework for semantic search and re-ranking using GPU-accelerated FAISS vector similarity to improve retrieval precision by 18%.
• Integrated LangChain for modular orchestration and LangGraph for designing composable retrieval–generation pipelines.
• Designed an API interface enabling plug-and-play model backends and dynamic query routing to optimize real-time responses.
• Deployed the system on AWS ECS using Docker, implementing CI/CD for environment reproducibility and continuous delivery.
• Optimized latency and GPU utilization by implementing caching and batched vector retrieval for high-throughput workloads.

EDUCATION

Master of Science, *Iowa State University* Aug 2023 – Aug 2025 | IA, USA
Computer Engineering (Computing and Networking Systems)
Bachelor of Technology, Amrita Vishwa Vidyapeetham
Electronics and Communication Aug 2016 – Aug 2020 | INDIA